Probabilistic Inventory models: Part 4

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Multi-period Model

- Used for inventory systems where **ordering, holding, and demand occur over multiple periods.**
- Unlike single-period models, these allow for **inventory carryover** between periods.

In multi-period model, no setup cost is assumed. Additionally, the model allows backlog of demand and assumes a zero-delivery lag. It further assumes that the demand D in any period is described by a stationary probability density function, pdf, $f(D)$.

Example

- In online platforms such as Amazon, Flipkart, there is no setup cost involved in stocking or reordering digital goods. Orders that cannot be fulfilled immediately are queued and delivered when available. Downloads are instant once available.
- For popular eBooks, demand stays statistically consistent over time

Multi-period Model

The multi-period model considers the discounted value of money. If $\alpha(< 1)$ is the discount factor per period, then an amount Rs. *A* available *n* periods from now has a present value (Rs.) of $\alpha^n A$.

Suppose that the inventory situation encompasses n periods and that unfilled demand can be backlogged exactly one period. Define

$F_i(x_i)$ = Maximum expected profit for periods i, $i + 1$, ... and n, given that x_i is the amount on hand before an order is placed in period i

Assuming that *c* and *r* are the cost and revenue per unit, respectively, the inventory situation can be formulated using

$$F_{n+1}\left(y_n - D\right) = 0$$

Multi-period Model

$$F_i(x_i) = \max_{y_i \geq x_i} \left[\begin{array}{l} -c(y_i - x_i) + \int_0^{y_i} \{rD - h(y_i - D)\} f(D)\,\mathrm{d}D \\ + \int_{y_i}^{\infty} \{ry_i + \alpha r(D - y_i) - p(D - y_i)\} f(D)\,\mathrm{d}D \\ + \alpha \int_0^{\infty} \{F_{i+1}(y_i - D) f(D)\,\mathrm{d}D\} \end{array} \right], \quad i = 1, 2, \ldots\ldots, n$$

- The value of x_i may be negative because unfilled demand is backlogged.
- The quantity $\alpha r(D - y_i)$ in the second integral is included because $(D - y_i)$ is the unfilled demand in period i that must be filled in period $i + 1$.

Multi-period Model

The problem can be solved recursively (involves repetition). For the case where the number of periods is infinite, the recursive equation reduces to

$$F(x) = \max_{y \geq x} \left[\begin{array}{l} -c(y-x) + \int_0^y \{rD - h(y-D)\} f(D)\,\mathrm{d}D \\ + \int_y^\infty \{ry + \alpha r(D-y) - p(D-y)\} f(D)\,\mathrm{d}D \\ + \alpha \int_0^\infty \{F(y-D)\} f(D)\,\mathrm{d}D \end{array} \right]$$

where *x* and *y* are the inventory levels for each period before and after an order is received, respectively.

Multi-period Model

The optimal value of y can be determined from the following necessary condition, which also happens to be sufficient because the expected revenue function F(x) is concave:

$$\frac{\partial(.)}{\partial y} = -c - h\int_0^y f(D)\,\mathrm{d}D + \int_y^\infty \{(1-\alpha)r + p\} f(D)\,\mathrm{d}D + \alpha\int_0^\infty \frac{\partial F(y-D)}{\partial y} f(D)\,\mathrm{d}D = 0$$

The value of $\frac{\partial F(y-D)}{\partial y}$ is determined as follows:

If there are $\beta(>0)$ more units on hand at the start of the next period, the profit for the next period will increase by $c\beta$, because this much less has to be ordered.

Multiperiod Model

This implies

$$\frac{\partial F(y-D)}{\partial y} = c$$

The necessary condition becomes

$$-c - h\int_0^y f(D)\,\mathrm{d}D + \{(1-\alpha)r + p\}\left\{1 - \int_0^y f(D)\,\mathrm{d}D\right\} + \alpha c\int_0^\infty f(D)\,\mathrm{d}D = 0$$

Multi-period Model

The optimum inventory level y^* is thus determined from

$$\int_0^{y^*} f(D)\,\mathrm{d}D = \frac{p+(1-\alpha)(r-c)}{p+h+(1-\alpha)r}$$

The optimal inventory policy for each period, given its entering inventory level x, is thus given as

If $x < y^*$, order y^*-x

If $x \geq y^*$, do not order.

A shopkeeper is selling umbrellas. Each umbrella costs ₹100 and is sold for ₹150. Any unsold umbrella is wasted. The demand is normally distributed with mean = 200 units and standard deviation = 50 units.

$$F(Q^*) = \frac{C_s}{C_s + C_o};$$

$$C_s = 50, C_o = 100.$$

Hence, critical ratio= 50/150=0.3333. For p=0.3333, Normsinv(0.3333) in MS Excel is gives $z=-0.43082$.

Order quantity=$200-0.43082\times50=178.459$; Hence, order 178 umbrellas.

A product's demand during lead time is normally distributed with **mean = 500 units** and **standard deviation = 50 units**. The company wants a **95% service level**. Find out the safety stock.

Normsinv(0.95) gives z=1.64485

- Mean = 500
- Std. Dev = 50
- Service level = 95% ⇒ z = 1.645

$$\text{Safety Stock} = z \cdot \sigma = 1.645 \cdot 50 = \boxed{82.25 \approx 82 \text{ units}}$$

Service Level

- In inventory management, service level refers to a company's ability to meet customer demand from available inventory.
- It's a critical metric for assessing how well a business fulfills orders promptly and accurately.
- A high service level means the company can consistently satisfy customer orders, while a low service level may lead to stockouts, backorders, and lost sales.

A company has a reorder point (ROP) policy. Lead time is 5 days. Daily demand is normally distributed with **mean = 100 units, std. deviation = 10 units**. The company wants a **90% service level**. What is the reorder point (ROP)?

- Daily demand mean = 100
- Std. dev = 10
- Lead time = 5 days
- Service level = 90% ⇒ z = 1.28

Step 1: Compute total mean demand during lead time

$$\mu_{LT} = 5 \cdot 100 = 500$$

Step 2: Compute standard deviation during lead time

$$\sigma_{LT} = \sqrt{5} \cdot 10 \approx 2.236 \cdot 10 = 22.36$$

Step 3: Reorder Point (ROP)

$$ROP = \mu_{LT} + z \cdot \sigma_{LT} = 500 + 1.28 \cdot 22.36 \approx 500 + 28.62 = \boxed{528.62 \approx 529 \text{ units}}$$

A distributor stocks 600 units of a product. The demand during lead time is **normally distributed**, with **mean = 550 units, std. dev = 40 units**. What is the probability of a **stock-out**?

- Stock = 600
- Mean demand = 550
- Std. dev = 40

Step 1: Calculate z-score

$$z = \frac{600 - 550}{40} = \frac{50}{40} = 1.25$$

Step 2: Stockout probability

$$P(\text{Stockout}) = P(Z > 1.25) = 1 - P(Z \leq 1.25)$$

From Z-table, P(Z ≤ 1.25) ≈ 0.8944

$$P(\text{Stockout}) = 1 - 0.8944 = \boxed{0.1056 \text{ or } 10.56\%}$$

A retailer wants to maintain a 95% fill rate. The standard deviation of demand during lead time is **20 units**, and the mean is **300 units**. How much inventory should be stocked to maintain this fill rate?

- Mean = 300
- Std. Dev = 20
- Fill rate = 95% ⇒ z = 1.645

$$\text{Inventory to stock} = \mu + z \cdot \sigma = 300 + 1.645 \cdot 20 = 300 + 32.9 = \boxed{332.9 \approx 333 \text{ units}}$$

Service Level

The probability **that no stockout occurs during the lead time.**

In other words, **how often** your inventory is **sufficient** to meet demand **without running out.**

Mathematically:

$$\text{Service Level } (\alpha) = P(\text{Demand during lead time} \leq \text{Reorder point})$$

Example:

If service level is 95%, then **in 95% of the cycles**, you will **not stock out** during lead time.

- It is **cycle-based**
- Used to **set safety stock**
- Doesn't tell how much demand was lost when a stockout occurs.

Fill rate

The proportion of **total demand that is actually filled from stock** on hand **without delay**.

Mathematically:

$$\text{Fill Rate } (\beta) = \frac{\text{Total units delivered from stock}}{\text{Total demand}}$$

Example:

If the fill rate is 95%, then **95% of units** requested by customers are **delivered immediately** from stock, and **5% are backordered or lost.**

- It is **unit-based**
- Takes into account **how much** demand was unmet during stockouts.